Pavan Kalyan Vottikundalu

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Career Objective

I'm an Electrical Engineering graduate student expected to graduate on June 2027 with hands-on experience in thin-film sputtering and wet chemical etching, built through thesis work on a fiber-optic hydrogen sensor. So I've learned a lot about how small changes in a recipe change the result, and how to keep going when a process doesn't work the first time. I'd like to bring that into a fab environment. And currently I'm looking for a process/equipment engineering internship or a full-time role after graduation.

Technical Expertise

- **Semiconductor Technology:** CMOS Fabrication, Photolithography
- **Process & Analysis:** SPC, DOE
- **Tools:** Python, JMP, Minitab, TCAD (Sentaurus), MATLAB
- **Programming:** Python, C, SQL

Education

Southern Taiwan University of Science and Technology *2025-present*
M.S. in Electrical Engineering
- Coursework: Device Physics, Solid-State Electronics, CMOS Technology
- Focus on MOSFET operation, short-channel effects, carrier transport, and device scaling

Sri Venkatesa Perumal College of Engineering and Technology *2019-2023*
B.Tech in Electronics & Communication Engineering
- First Class Degree
- Coursework: VLSI Design, Digital Electronics, Analog Circuits, Semiconductor Devices

Experience

Junior Software Engineer, Trinite Digital — Pondicherry, India
- Developed responsive user interfaces using HTML, CSS, JavaScript, and WordPress
- Optimized performance for both mobile and desktop platforms

Lab & Process Experience

- I run both RF and DC magnetron sputtering to deposit thin films.
- Wet etching fibers with HF and checking results under microscope and optical power measurement.

Projects

Optical Hydrogen Sensor Development — WO_3+Pd Coated Glass Optical Fiber *STUST | 2026 – Present*
- Coating an optical fiber with two thin layers Wo3+Pd by sputtering, to build a hydrogen sensor.
- Etching fibers down to a precise 9μm diameter and checking the results under a microscope and with Optical power readings.

Matrix AI — Bayesian Optimization Platform for Process Development
- Built and deployed a web app (FastAPI, scikit-learn, PostgreSQL) that reduced required sputtering experiments from 72 to an estimated 15–20.
- It uses a Gaussian process model with an acquisition function that balances multiple process goals at once.